

Answer the following questions:

1. Which of the following system calls is typically **not** used in UDP communication?

- a) bind()
- b) connect()
- c) socket()
- d) close()

Answer: connect()

2. Which of the following system calls is used to establish a connection between a client and a server in a TCP based connection?

- a) fork()
- b) bind()
- c) connect()
- d) write()

Answer: connect()

3. State True or False. In connection-less communication, each message carries the destination address and is routed independently from source to the destination.

Answer: True

4. State True or False. Concatenation of IP address, port and process name defines a socket.

Answer: False

5. State True or False. Network communication protocols typically use big-endian byte order.

Answer: True

6. In a Listen system call, what does the parameter Backlog specify?

- a) maximum number of pending connections allowed
- b) maximum number of concurrent connections allowed
- c) minimum number of concurrent connections allowed
- d) minimum number of pending connections allowed

Answer: a)

7. Complete the code snippet for establishing a TCP socket at a server. There are five statements that are to be completed (text in red color indicates where the code is to be inserted):

```
int sockfd, bindret, fd, clientfd;

    socklen_t clilen;
    struct sockaddr_in serv_addr, cli_addr;
    printf("Starting server...\n");
    sockfd = socket(AF_INET, ,0); /* 1: What will be the second argument
of the socket system call to create a TCP socket */

    serv_addr.sin_family = ; /* 2: What will be the value that needs to
be given to this field if the address family is IPV4*/
```

```
serv_addr.sin_addr.s_addr =    ; /* 3: What should be the value of
this field if we wish to bind all local interfaces of the host*/

serv_addr.sin_port =    ; /*4: What should be the value of this field
if we want to use port no 2345 as the server port*/
[D]
bindret = bind(sockfd, ,sizeof(serv_addr)); /* 5 : What will be the
second argument of bind() system call here*/
```

Answer:

1. SOCK_STREAM
2. AF_INET (Case Sensitive)
3. INADDR_ANY (Case Sensitive)
4. htons(2345)
5. (struct sockaddr *)&serv_addr

8. What is the fundamental difference between a program and a process?
- a) A program is a passive entity; a process is an active entity
 - b) A program is an active entity; a process is a passive entity
 - c) Programs require CPU, memory, and I/O; processes do not require any resources
 - d) There is no practical difference; the terms are interchangeable

Answer: a)

9. In the process lifecycle, which of the following transitions is NOT possible directly?
- a) From running to waiting
 - b) From ready to terminated
 - c) From waiting to ready
 - d) From new to ready

Answer: b)

10. Which of the following best describes independent processes?
- a) Processes that share memory and communicate frequently
 - b) Processes that do not interact with other processes and do not affect them
 - c) Processes that are created by the same parent process
 - d) Processes that run on different processors

Answer: b)

11. In message passing IPC, which two primitive operations MUST be provided by the system?
- a) fork() and exec()
 - b) send() and receive()
 - c) bind() and connect()
 - d) listen() and accept()

Answer: b)

12. A socket endpoint for communication requires which of the following information?

- a) Only the IP address of the host
- b) IP address, port number, and process name
- c) IP address and port number
- d) Port number, protocol type, and hostname

Answer: c)

13. Port numbers 0-1023 are reserved and require special privileges. Which of these ports is correctly matched with its service?

- a) Port 80 is for FTP
- b) Port 23 is for HTTP
- c) Port 21 is for Telnet
- d) Port 80 is for HTTP

Answer: d)

14. Why is TCP generally slower than UDP?

- a) TCP uses more network bandwidth
- b) TCP uses a larger packet size
- c) TCP requires connection establishment, acknowledgments, and retransmission for reliability
- d) TCP transmits data in smaller chunks

Answer: c)

15. In a blocking (synchronous) send operation, what does the sender do after initiating a send operation?

- a) Continues executing immediately
- b) Waits until the send operation completes before continuing
- c) Sends the message and periodically checks whether it has been delivered
- d) Delegates the send operation to another process and continues immediately

Answer: b)

16. Which of the following is TRUE about UDP client-server communication?

- a) The client must call `listen()` before sending data
- b) The server must call `accept()` to establish a connection
- c) Both client and server use `sendto()` and `recvfrom()` to exchange data
- d) UDP requires a three-way handshake like TCP

Answer: c)

17. What is the key difference between blocking send and blocking receive in message passing?

- a) Send is always faster than receive
- b) Send waits for the send operation to complete; receive waits for a message to become available
- c) Send never blocks; only receive blocks
- d) There is no practical difference between them

Answer: b)

18. Which IPC model is typically better for processes on different machines connected by a network?
- a) Shared memory
 - b) Message passing
 - c) Both are equally suitable
 - d) Neither model works over networks

Answer: b)

19. What is the key difference between SOCK_STREAM and SOCK_DGRAM socket types?
- a) SOCK_STREAM is faster than SOCK_DGRAM
 - b) SOCK_STREAM provides reliable, connection-oriented communication; SOCK_DGRAM provides unreliable, connectionless communication
 - c) SOCK_STREAM is used for UDP; SOCK_DGRAM is used for TCP
 - d) Both are identical in functionality but have different names

Answer: b)

20. In a typical TCP server implementation, after `accept()` returns successfully, what does the returned file descriptor represent?
- a) The listening socket waiting for more connections
 - b) A new socket connected to the specific client that just connected
 - c) The server's own communication endpoint
 - d) A file on the server's filesystem

Answer: b)